Solutions to Advanced Problems by Stephen Siklos

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Problem 1

(i) Find all sets of positive integers a, b and c that satisfy the equation

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1.$$

(ii) Determine the sets of positive integers a, b and c that satisfy the inequality

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 1.$$

Solution to Problem 1

For part (i), the equation is symmetric so, without loss of generality, assume that $c \leq b \leq a$. Multiply through by abc, to get $ab + bc + ca = abc$. We claim that $c \leq 3$. Here is the proof, Suppose, for the sake of contradiction that $c > 3$, then $3ab < abc = ab + bc + ca \leq ab + ac + ab$ and so $ab < ac \iff b < c$ which is a contradiction since we assumed $b \geq c$. Now we can just check each case.

Case 1: $c = 1$. Then $ab + b + a = ab \iff a + b = 0$ but since $a, b \geq 1$ this is impossible.

Case 2: $c = 2$. Then $ab + 2(a + b) = 2ab \iff 2(a + b) = ab \iff 2b = ab - 2a \iff a = \frac{2b}{b-2}$. Since a is a positive integer we need $2b \equiv (b - 2) + (b - 2) + 4 \equiv 4 \equiv 0 \pmod{b - 2}$. Thus $b - 2$ must be a factor of 4. We must have $b - 2 = 1, 2, 4$ and so $b = 3, 4, 6$. When $b = 6$, $a = \frac{12}{4} = 3$ which is impossible since $a \geq b$. When $b = 4$, $a = \frac{8}{2} = 4$. When $b = 3$, $a = \frac{6}{1}$. So the solutions here are $(a, b, c) = (4, 4, 2), (6, 3, 2)$.

Case 3: $c = 3$. Then $ab + 3(a + b) = 3ab \iff 3(a + b) = 2ab \iff 3b = 2ab - 3a \iff a = \frac{3b}{2b-3}$. Since a is a positive integer we need $3b \equiv b + 3 \equiv 0 \pmod{2b - 3}$. Notice that if $2b - 3 > b + 3 \iff b > 6$, then we must have $b + 3 = 0 \iff b = -3$ since it can't be a positive multiple of $2b + 3$. But since b is positive this is impossible, so $3 = c \leq b \leq 6$. Now just try each value of b. When $b = 3$, $a = \frac{9}{3} = 3$. When $b = 4$, $a = \frac{12}{5} \notin \mathbb{N}$, so this is impossible. When $b = 5$, $a = \frac{15}{7} \notin \mathbb{N}$, so this is impossible. When $b = 6$, $a = \frac{18}{9} = 2$ but since $a \geq b$ this is impossible. So the only solution here is $(a, b, c) = (3, 3, 3)$.

Finally combining each of the three cases shows that the only solutions are $(a, b, c) = (4, 4, 2), (6, 3, 2)$ and $(3, 3, 3)$ up to permutation.

Part (ii) follows similarly, suppose that $c \geq 3$, then $\frac{1}{a} \leq \frac{1}{b} \leq \frac{1}{c} \leq \frac{1}{3}$ but this forces the equality case $a = b = c = 3$. Now what if $c = 2$, then $\frac{1}{a} + \frac{1}{b} \geq \frac{1}{2}$. We must have $b \leq 4$, because if $b > 5$ then $\frac{1}{5} > \frac{1}{b} \geq \frac{1}{a}$ and there is no way that they can sum to more than $\frac{1}{2}$. When $b = 2$, a can be any natural number. When $b = 3$, $\frac{1}{a} \geq \frac{1}{6} \iff a \leq 6$. Now what if $c = 1$, then a, b can be any natural numbers.